+49 176 86075720 serhatkumas@outlook.com Passau, DE

Serhat Kumas

Solutions Architect

AI-focused Solutions Architect and Software Engineer with a Master's in Computer Science and strong field experience in designing intelligent, scalable systems. Proven track record in practical AI-driven development, specializing in complex NLP integrations and microservices architecture. Actively expanding expertise in core AI fundamentals to continuously bridge innovative machine learning capabilities with robust backend engineering, delivering next-generation enterprise applications. Beyond system architecture, a dedicated community organizer who initiated the Google Developer Group (GDG) on Campus Passau to foster local developer ecosystems and technical collaboration.

Experience

Solution Architect BSH Hausgeräte (Munich) Jan 2026 - Present

- Architected and implemented a full-stack, AWS-deployed video management platform (React.js, Node.js) featuring SSO and OAuth for multi-channel batch uploading, CRUD operations, and analytics, successfully securing an enterprise YouTube API quota increase.
- Developed targeted no-code AI solutions featuring stringent safety guardrails and robust context-feeding mechanisms, including a company-wide help desk agent and a specialized code-helper agent for non-technical staff.
- Engineered a scalable, hierarchical multi-agent architecture in Python using the Google Agent Development Kit (ADK), implementing Agent-to-Agent (A2A) protocols, dynamic intent routing, and advanced contextual guardrails.

Software Developer DC Academy Mar 2022 - Feb 2023

- Engineered a Python/Flask REST API integrating Google Dialogflow CX to process NLP requests and facilitate enterprise data exchange.
- Built an interactive web UI (HTML, CSS, JavaScript ES6+) that asynchronously consumes backend APIs for real-time chatbot interactions.
- Integrated the ChatGPT API within a microservices architecture to asynchronously process complex queries and elevate NLP capabilities.

Intern Game Developer Spike Games Jan 2022 - Feb 2022

- Engineered core 2D game mechanics and systems within the Unity engine, writing performant C# code to implement features such as a state-driven character controller, event-based interaction systems, and a customizable inventory management module.
- Architected robust C# scripts to control complex character animation state machines and finite state machine (FSM) based enemy behavior.

Intern Software Developer DC Academy Nov 2021 - Jan 2022

- Acquired foundational knowledge of blockchain development by building and testing basic smart contracts in Solidity, gaining hands-on experience with key concepts like transaction execution and gas optimization.
- Spearheaded the design of service logic and user flow architecture for a ride-sharing application, modeling core systems such as driver-passenger matching algorithms, payment processing integration, and trip status management.

Intern Backend Developer Delta Smart Technologies Jan 2021 - Feb 2021

- Built Spring Boot REST APIs and managed data persistence via Spring Data JPA for standard CRUD operations, applying strict layered architecture principles to reduce code complexity, minimize bugs, and improve overall maintainability. To further support these systems, optimized SQL queries and stored procedures, driving high-performance data access and integrity across Java and Python backend services.

Off-Field Experience

Area Supervisor High Sierra Pools (USA) May 2024 - Sep 2024

- Directed workflows for a 30-person team and served as the primary contact for 12 clients, optimizing operational efficiency and driving client satisfaction.



Education

Universität Passau

M.Sc. in Computer Science
2025-Current

Izmir Institute Of Technology

B.Sc. in Computer Engineering
2018-2023

Anadolu University

B.Sc. in Business Administration
2020-2025

Skills

- Techs : Python, Java, C#, JavaScript, PostgreSQL, HTML, CSS, SQL
- Frameworks: Flask, Spring Boot, Unity
- Concepts: OOP, RESTful API, Smart Contracts, Distributed Systems, Algorithms, Scrum, Project Management, Data Structures, OAuth, SSO
- Tools: Git, Jira, Docker, Postman, AWS

Language

Turkish

Native

English

C1 - Advanced

German

A2 - Beginner (Still Learning)

Portfolio

Linkedin
https://www.linkedin.com/in/serhatkumas/

Github
https://github.com/SerhatKumas

Portfolio Website
https://serhatkumas.github.io